# D$^2$M-DETR: Differential Decoupling and Multiscale Dynamic Modeling for Remote Sensing Small Object Detection

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***Abstract*—In recent years, small object detection in remote sensing images has been widely applied in many fields, yet it still faces several serious challenges. On one hand, small objects inherently have weak features and are easily disturbed by complex backgrounds. On the other hand, small objects in remote sensing images are typically densely distributed and often accompanied by occlusion, which easily leads to missed detections and false positives. To tackle these challenges, we introduce an end-to-end detection network named D$^2$M-DETR, which integrates differential decoupling and multiscale dynamic modeling. The overall network comprises the differential decoupled salient features (DDSFs) strategy and the multiscale dynamic spatial context modeling (MDSCM) module. The DDSF strategy employs a dual-branch backbone for global and local feature extraction. It then applies differential decoupling to separate and enhance unique and basic features, effectively suppressing background interference and improving the discriminability of small objects. The MDSCM module jointly constructs multiscale dynamic local focus and multidirectional global spatial modeling, and enhances interactions among multiscale contextual information. This enables explicit modeling of structural priors and semantic relationships of small objects in densely occluded scenes. Our D$^2$M-DETR achieves excellent detection performance on multiple remote sensing datasets, validating the efficacy of the proposed algorithm. The source code is available at https://github.com/Xidian-AIGroup190726/DDMDETR**

***Index Terms*—Detection transformer (DETR), differential decoupling, multiscale dynamic modeling, remote sensing object detection, small objects.**

## I. Introduction

REMOTE sensing object detection has long held a significant position in the domain of remote sensing. With its core lying in the precise localization and identification of objects, it plays an irreplaceable role in critical domains such as military reconnaissance, environmental monitoring, urban planning, and disaster assessment, [1], [2]. However, remote sensing data is growing exponentially, and application scenarios are becoming increasingly complex. The key challenge now is how to effectively enhance the detection accuracy of remote sensing objects in complex scenarios. These conditions include severe background interference, diverse object distributions, and limited feature representation.



(a) (b) (c)

Fig. 1. Complex scenes in remote sensing images. (a) Complex background interference for weak small objects. (b) and (c) Dense distribution and occlusion for small objects.

In recent years, computational resources and deep neural networks have advanced significantly [3], [4]. These advances have enabled deep learning-based object detection to extract image features adaptively. As a result, detection accuracy and model generalization in complex environments have improved substantially [5]. These detection methods can be primarily categorized into the following four types based on their core design principles: single-stage methods, two-stage methods, anchor-free methods, and transformer-based methods.

Single-stage methods, represented by the YOLO series [6] and SSD series [7], directly achieve object localization and classification through dense prediction. Two-stage methods, represented by the R-CNN series [8], [9], and SPPNet [10], these approaches first extract region proposals followed by classification and regression. Anchor-free methods, exemplified by FCOS [11], CenterNet [12], and LCP [13], eliminate the anchor mechanism by predicting keypoints or center points, leading to a simpler architecture. Transformer-based methods, represented by detection transformer (DETR) [14], deformable-DETR [15], and DINO [16], treat object detection as a set prediction problem. They leverage the transformer encoder–decoder architecture with a bipartite matching loss to achieve truly end-to-end detection, thereby removing the

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reliance on handcrafted components like anchor boxes or nonmaximum suppression (NMS).

However, the methods above are primarily based on research using natural image datasets and often exhibit limited performance when applied to remote sensing images. Owing to their unique imaging mechanism, which is characterized by a top–down perspective, high resolution, and extensive coverage, remote sensing images frequently contain dense small objects. As illustrated in Fig. 1, compared to natural images, complex remote sensing images present two major outstanding challenges for object detection.

1) Small objects have intrinsically weak features, making them prone to interference from noise and similar objects in complex backgrounds. This results in low distinguishability between target and background, thereby exacerbating the detection challenge.
2) Small objects in remote sensing scenes are typically densely distributed and highly susceptible to severe occlusion. These characteristics frequently result in missed detections and false positives, which significantly degrade detection performance.

To address the first challenge, it is crucial to suppress background interference and enhance small object detection. Zhu et al. [17] enhance small object features by integrating global context and effectively suppress background interference through dynamic aggregation of key region information. Zhang et al. [18] introduced edge prior information to construct a dual-branch network for feature-edge interactive enhancement, and designed a progressive multiscale fusion mechanism to preserve small object details in deep features while suppressing background interference. Li et al. [19] enhance multilevel feature interaction through a bidirectional triangular cyclic fusion structure and bridge the multiscale semantic gap with a hybrid dual-attention module. They also efficiently aggregate multiscale features using a reparameterized bidirectional fusion module. Despite the encouraging results achieved by the methods above, most fail to adequately account for the difference between small objects and the background at both global and local feature levels. Global features tend to over-respond to the background, overwhelming small objects. Although local features can capture fine details, their limited receptive fields make it difficult to distinguish background interference with similar textures. Consequently, they struggle to suppress background interference effectively.

To address the second challenge, the problem of dense distribution and occlusion of small objects, researchers have also proposed many methods to solve it. Song et al. [20] effectively enhanced the feature representation capability of small objects in dense and occluded scenes by dynamically adjusting spatial reference points and adaptively fusing multiscale features. Wang et al. [21] improved small object detection in densely packed and occluded scenes by holistically fusing correlation information from multiple scales, spatial regions, and hierarchical levels within images. This integration enhances the salient features of small objects while mitigating background interference. Yang et al. [22] achieved efficient and robust small object detection in complex scenes through dynamic multiscale aggregation, cross-layer context enhancement, low-dimensional feature reuse, and a progressively simplified framework. Recent studies further improve tiny-object localization and representation through distribution-based bounding-box modeling, coarse-to-fine proposal generation, and imitation learning [23], [24]. However, most existing methods primarily focus on enhancing feature representation and leveraging contextual information for small objects in complex scenes. While they have achieved notable progress in these aspects, they generally fail to explicitly model the semantic relationships and structural priors among densely distributed and occluded objects, particularly for irregular small targets, where occlusion and dense distribution disrupt appearance and spatial layout.

Mamba [25] offers strong potential for addressing this limitation by efficiently modeling long-range dependencies and spatial correlations among densely distributed small objects. Recent vision Mamba methods have further extended this capability to visual tasks through multidirectional scanning strategies. However, directly applying existing vision Mamba structures to remote sensing small object detection remains limited, especially in dense and occluded scenarios. Most of these methods primarily focus on global spatial modeling, often neglecting multiscale local key regions, irregular object layouts, and cross-scale contextual interactions.

In response to these challenges, we propose the differential decoupling and multiscale dynamic modeling network ($D^2$M-DETR). The network primarily consists of two components: the differential decoupled salient features (DDSFs) strategy and the multiscale dynamic spatial context modeling (MDSCM) module.

It is noteworthy that this study adopts an absolute size criterion for defining small objects, where any object with a pixel area less than $32 \times 32$ is considered a small object [26]. This criterion is used only for dataset statistics and $AP_s$ evaluation; it does not impose a hard size constraint on the network architecture or inference process.

To summarize, our main contributions are as follows.

1) We propose a DDSFs strategy, which decouples and enhances the unique and basic features of global–local through a dual-branch backbone. Combined with a residual fusion mechanism, it significantly improves the discriminability of small objects and suppresses background interference.
2) We propose an MDSCM module. It uses three key components: multiscale deformable attention (MSDeformAttn), a four-directional cross-selective scan, and a gated aggregation stream. These components work together to build dynamic local focus at multiple scales, model global spatial context from various directions, and enhance interaction among multiscale contextual information. This design explicitly captures the structural priors and semantic relationships of small objects, thereby effectively improving their detection performance in dense and occluded scenarios.

The rest of this article is structured as follows. Section II presents a review of related work. Section III provides a detailed description of the proposed method. Section IV presents the experimental results, including our experimental

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data, ablation studies, and comparative experiments. Finally, Section V concludes this work.

## II. Related Work

### A. CNN and Transformer Networks

*1) CNN Networks:* In recent years, convolutional neural networks have remained indispensable in tasks such as object detection. ResNet [27] addressed the vanishing gradient problem in deep neural networks by introducing a residual learning framework with cross-layer skip connections, and has since been extensively adopted. DenseNet [28] improved network parameter efficiency and depth through a dense connection mechanism and feature reuse design. EfficientNet [29] balanced accuracy and efficiency through composite scaling methods and NAS technology. ConvNeXt [30] revisits the design of the vision transformer and proposes a $7 \times 7$ deep convolutional backbone, significantly improving detection performance. LSKNet [31] demonstrates the advantages of large convolutional kernels in remote sensing detection through large kernel selection modules and dynamic receptive field mechanisms, particularly in small object scenarios. Subsequently, pure CNN designs increasingly favor large receptive fields to achieve strong competitiveness.

*2) Transformer Networks:* The transformer, originally proposed in natural language processing with great success, was later adapted to computer vision, where its self-attention mechanism captures global, long-range image dependencies. The vision transformer [32] first demonstrated the viability of pure Transformers in vision via fixed patch partitioning and global self-attention. DeiT [33] introduced a token distillation mechanism and attention distillation strategy, effectively reducing data dependency. Swin transformer (SwinT) [34] enables global modeling at linear complexity via hierarchical shifted-window self-attention. Building on this, CSWin transformer [35] enhances local–global fusion via cross-shaped window self-attention and local-enhanced position encoding. The transformer has made remarkable strides in computer vision owing to its strong global modeling capacity, leading to widespread enhancements and applications across diverse networks and tasks [36], [37].

*3) CNN-Transformer Networks:* To fully leverage the complementary relationship between CNNs and transformers, recent works have demonstrated that combining the advantages of CNNs in handling bias and transformers in handling long-term dependencies can be achieved through hybrid architectures. ConFormer [38] integrates local and global features at different resolutions through a parallel dual-branch structure and an interactive feature fusion mechanism. GhostFormer [39] fuses cross-modal information via hierarchical interactions, combining depthwise convolutions with sparse attention. RingMo-Lite [40] enhances lightweight remote sensing models via a hybrid CNN-transformer and frequency-domain masked pretraining. FBDPN [41] boosts small object detection and background suppression via a CNN-transformer hybrid and differential pyramid, leveraging dual-path enhancement and multiscale context modeling. By combining the strengths of CNNs and transformers, the hybrid architecture synergistically boosts the model's capacity to capture local details and reason about global context, presenting a promising solution for tackling small object detection in complex remote sensing environments. Unlike the aforementioned CNN–transformer hybrid methods, which primarily exchange or aggregate local and global features, DDSF explicitly exploits branch response differences to decouple and enhance unique and basic features before residual fusion.

### B. Vision Mamba

State-space models (SSMs) are linear time-invariant systems. The system stores contextual information in its internal hidden state $h(t) \in \mathbb{R}^N$, and employs this to transform the input $x(t) \in \mathbb{R}$ into the output $y(t) \in \mathbb{R}$. Its computational procedure is as follows:

$$h'(t) = \boldsymbol{A}h(t) + \boldsymbol{B}x(t), \quad y(t) = \boldsymbol{C}h(t) + Dx(t) \tag{1}$$

where $\boldsymbol{A} \in \mathbb{R}^{N\times N}$ denotes the state matrix, $\boldsymbol{B} \in \mathbb{R}^{N\times 1}$ the input matrix, $\boldsymbol{C} \in \mathbb{R}^{1\times N}$ the output matrix, and $D \in \mathbb{R}$ the feedthrough (or direct transmission) matrix. To integrate SSMs into deep networks, discretize continuous parameters via ZOH and rewrite the SSM as follows:

$$\boldsymbol{h_i} = \overline{\boldsymbol{A}}h_{i-1} + \overline{\boldsymbol{B}}x_i, y_i = \boldsymbol{C}h_i + D\ x_i \tag{2}$$

$$\overline{\boldsymbol{A}} = \exp(\Delta\boldsymbol{A}), \quad \overline{\boldsymbol{B}} = (\Delta\boldsymbol{A})^{-1}\left(\exp(\Delta\boldsymbol{A}) - \boldsymbol{I}\right)\cdot\Delta\boldsymbol{B} \tag{3}$$

where the continuous parameters $\boldsymbol{A}$ and $\boldsymbol{B}$ are discretized into $\overline{\boldsymbol{A}} \in \mathbb{R}^{N\times N}$ and $\overline{\boldsymbol{B}} \in \mathbb{R}^{N\times 1}$ using a timescale parameter $\Delta \in \mathbb{R}$.

SSMs have limitations in modeling long-range dependencies. To tackle this issue, Mamba [25] proposed the selective SSM. This model uses hardware-aware algorithms and dynamically adjusts input-dependent parameters. As a result, it is able to model long-range dependencies efficiently with linear computational complexity. It has rapidly expanded into the field of computer vision, becoming an important foundational architecture. Vim [42] innovatively combines a bidirectional SSM with positional embeddings, becoming the first to migrate the Mamba architecture to the visual domain efficiently. It addresses the one-way modeling issue and the lack of positional awareness in Mamba when handling image sequences. VMamba [43] proposes a cross-scanning strategy, enabling natural traversal of the 2-D image plane and capturing the inherent 2-D spatial relationships within images. Subsequent research focused on the scanning characteristics of image data. Researchers designed various scanning methods, including continuous 2-D scanning [44], diagonal scanning [45], local scan [46], and K-space circular scanning [47]. Each of these methods shows unique advantages in different application scenarios.

Mamba's performance in visual tasks exhibits task dependency. MambaOut [48] experimentally demonstrated that the structured SSM is not essential for image classification tasks, as performance remains excellent even after its removal. However, SSM is highly valuable in tasks requiring long-term sequence dependency modeling, particularly in the field of remote sensing object detection. MV-YOLO [49] innovatively integrates Mamba with self-attention mechanisms to optimize the detection of small, dense objects in remote sensing. ME-MSA-YOLO [50] effectively integrates global context and

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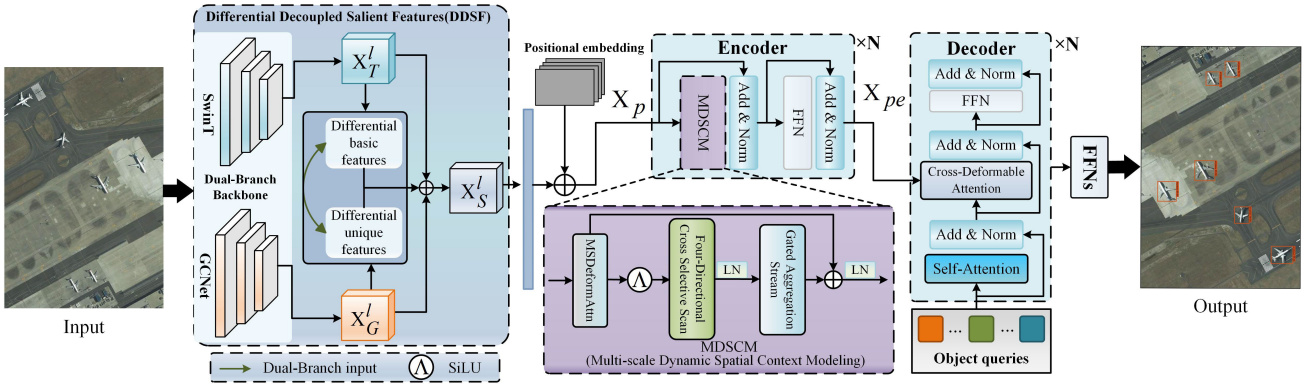



Fig. 2. Overall framework of the proposed D$^2$M-DETR begins by feeding the input image into a dual-branch backbone that combines SwinT and GCNet to extract features. The features then undergo differential decoupling processing and fusion to suppress background interference. Multiscale features combined with positional encoding are fed into an encoder, where multiscale dynamic context modeling enhances small object detection capability. Through iterative optimization, the decoder generates a fixed set of predictions in parallel, and finally, the object categories and bounding boxes are output by FFNs.

multiscale local features through Mamba and multiscale spatial attention mechanisms, markedly enhancing the accuracy of small object detection in remote sensing imagery.

Thanks to its linear computational complexity, Mamba strikes an optimal balance between computational efficiency and global context modeling in long-sequence tasks, such as processing high-resolution remote sensing images. It uses a scanning strategy combined with multidirectional sequence modeling. This further enhances spatial continuity by efficiently capturing long-distance dependencies and detailed features. As a result, it offers another solution for remote sensing of small objects.

## III. Method

This section presents a comprehensive overview of the proposed D$^2$M-DETR. First, we outline its overall architecture. Subsequently, a DDSFs strategy is proposed. Finally, we introduce the MDSCM module we designed.

### A. Overall Architecture

Fig. 2 illustrates the overall architecture of the proposed D$^2$M-DETR, which mainly consists of the following components.

*1) DDSF Strategy:* We begin by employing a parallel dual-branch backbone to extract features from the input image, obtaining feature representations $\mathrm{F}_i^T$ and $\mathrm{F}_i^G$, respectively. They are then separately passed into the differential unique features module and the differential basic features module, where differential decoupling is applied to produce enhanced features. Finally, the multifeature fusion is performed to obtain the salient feature $\mathrm{F}_i^S$. This process is defined as follows:

$$\mathrm{X}_S^l = \mathrm{DDSF}\left(\mathrm{X}_T^l, \mathrm{X}_G^l\right), \quad l \in \{1, 2, 3\} \tag{4}$$

where $i$ denotes the multiscale features, and this strategy effectively suppresses interference from complex backgrounds while highlighting the detailed features of small objects. See Section III-B for details. Then, convolutional operations are used to construct four-level multiscale feature maps with identical channels, which are then added with positional encodings and fed into the encoder.

*2) Encoder:* First, the multiscale feature maps are passed through the MDSCM module for local–global spatial modeling and multiscale contextual feature interaction. Subsequently, the features are further processed through a feed-forward neural network. Throughout the entire process, residual connections and layer normalization are employed to ensure training stability. Finally, the encoder structure is constructed by stacking N of the modules above. This process is defined as follows:

$$\mathbf{X}_{pe} = \mathcal{E}_{\mathrm{MDSCM}}\left(\mathbf{X}_p; \{\mathbf{x}^l\}_{l=1}^L\right) \tag{5}$$

where $\mathcal{E}_{\mathrm{MDSCM}}$ represents the encoder mapping function of the entire MDSCM module, $\mathbf{X}p$ is the input sequence, and $\{\mathbf{x}^l\}_{l=1}^L$ is the set of multiscale feature maps. See Section III-C for details.

*3) Decoding and Set Prediction:* First, the learnable object queries and the multiscale feature maps from the encoder are input together into the decoder. Subsequently, query interaction and feature decoding are performed through self-attention and deformable cross-attention mechanisms, progressively refining the object representations. Finally, the categories and bounding boxes for all queries are output in parallel through shared feed-forward networks (FFNs), directly forming the final prediction set

$$\mathrm{L}_{\mathrm{total}} = \lambda_{\mathrm{class}} \cdot \mathrm{L}_{\mathrm{class}} + \lambda_{\mathrm{box}} \cdot \mathrm{L}_{\mathrm{box}} + \lambda_{\mathrm{giou}} \cdot \mathrm{L}_{\mathrm{giou}} \tag{6}$$

where $L_{\mathrm{class}}$, $L_{\mathrm{box}}$, and $L_{\mathrm{giou}}$ represent the classification loss, the L1 loss for bounding box regression, and the GIoU loss, respectively.

### B. Differential Decoupled Salient Features

In remote sensing images, small objects possess limited features, and their scarce visual information is easily overwhelmed by cluttered and complex background noise. To alleviate this issue, we propose the DDSF strategy, which utilizes a dual-branch structure to separate and enhance the unique and basic features of objects. The strategy consists of four steps: dual-branch backbone, differential unique features, differential basic features, and feature fusion.

Here, unique features emphasize complementary details revealed by inconsistent global–local responses, whereas basic

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features preserve object information consistently supported by both branches while suppressing branch-dependent background interference.

*1) Dual-Branch Backbone:* In remote sensing images, objects of different scales within complex environments require differentiated contextual information, necessitating adaptive receptive fields. Since a single network architecture struggles to model both local details and global dependencies simultaneously, we adopt a dual-branch backbone that leverages transformers and CNNs for global and local feature extraction, respectively.

Given the high-resolution characteristics of remote sensing images, we employ the SwinT [34] as the feature extractor in the transformer branch

$$\begin{aligned} \hat{x}^{l} &= \text{W-MSA}\left(\text{LN}\left(x^{l-1}\right)\right) + x^{l-1} \\ x^{l} &= \text{MLP}\left(\text{LN}\left(\hat{x}^{l}\right)\right) + \hat{x}^{l} \\ \hat{x}^{l+1} &= \text{SW-MSA}\left(\text{LN}\left(x^{l}\right)\right) + x^{l} \\ x^{l+1} &= \text{MLP}\left(\text{LN}\left(\hat{x}^{l+1}\right)\right) + \hat{x}^{l+1} \end{aligned} \tag{7}$$

where $\hat{x}^{l}$ and $x^{l}$ represent the output features of the W-MSA and MLP of the $l$th module, respectively, and similarly for $\hat{x}^{l+1}$ and $x^{l+1}$. W-MSA refers to multihead self-attention computed within fixed windows, whereas SW-MSA refers to multihead self-attention applied over windows that are shifted across adjacent partitions. LN denotes layer normalization, and MLP stands for multilayer perceptron.

The SwinT employs hierarchical local and shifted window self-attention for efficient multiscale feature extraction, global context modeling, and significantly reduced computational complexity.

For the CNN branch, we employ an adapted GCNet network. To minimize the impact of architectural design and highlight the core characteristics of CNN and transformer, we represent the architecture as follows:

$$\begin{aligned} \hat{x}^{l} &= \text{GCB}\left(\text{LN}\left(x^{l-1}\right)\right) + x^{l-1} \\ x^{l} &= \text{MLP}\left(\text{LN}\left(\hat{x}^{l}\right)\right) + \hat{x}^{l} \end{aligned} \tag{8}$$

where $\tilde{x}^{l}$ and $x^{l}$ represent the output features of the GCB and GCNet of the $l$th module, respectively. LN denotes layer normalization, and MLP stands for multilayer perceptron.

Meanwhile, we design the GCB to reduce its structural discrepancy with the multihead self-attention mechanism. This enables the GCB to extract information from multiple subspaces through a multihead mechanism. The detailed design is described as follows:

$$\text{GCB} = \text{Concat}\Big(GC_1\left(x_1\right), \ldots, GC_g\left(x_g\right)\Big)W^{P}. \tag{9}$$

Here, the GCB divides the channels into $g$ groups through group convolution, where GC represents the convolution operation conducted independently within each group. $W^{P}$ is a pointwise convolution that acts as a linear transformation to enable feature interaction between different groups. In microdesign, we employ layer normalization and the GELU activation function. Moreover, to ensure consistency between the dual-branch structures, the number of blocks in each stage of GCNet is configured identically to that of SwinT.

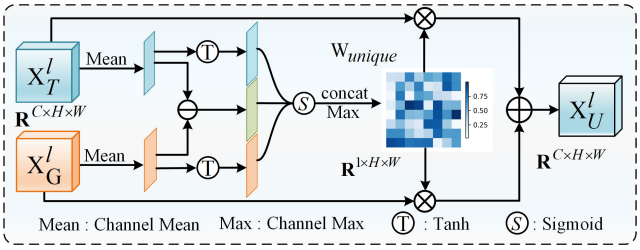



Fig. 3. Proposed differential unique features module. Derive unique information by differential computation, selectively enhance the response points, and yield the differential unique feature $\mathrm{X}_i^U$.

The design of GCNet implies local spatial inductive bias, which complements the global dependency modeling capability of the SwinT. Additionally, it serves as the input for the subsequent difference decoupling operation.

*2) Differential Unique Features:* Since the features of the SwinT branch are global in nature and those of the GCNet branch are local, detailed features in an image elicit different response levels from them. Therefore, we handle the differences in detailed features with the following specific operations. The architecture of the module is illustrated in Fig. 3.

To effectively handle detailed features, we first reduce channel redundancy, retain essential spatial information, and mitigate the impact of interchannel outliers. The formula is as follows:

$$\mathrm{X}_i^{Tm} = \text{Mean}\left(\mathrm{X}_i^{T}\right),\ \mathrm{X}_i^{Gm} = \text{Mean}\left(\mathrm{X}_i^{G}\right) \tag{10}$$

where Mean(·) normalizes the channel mean in the spatial dimension to obtain the fundamental spatial features $\mathrm{X}_i^{Tm} \in \mathbb{R}^{1\times H_i\times W_i}$ and $\mathrm{X}_i^{Gm} \in \mathbb{R}^{1\times H_i\times W_i}$. The spatial characteristics of both can be divided into four categories of different response information: high-high, low-low, low-high, and high-low responses. The greater the disparity between low-to-high and high-to-low responses, the more likely it represents a controversial, detailed feature. Therefore, we calculate the difference feature $\mathrm{X}_i^{D1}$ by subtraction, which is expressed by the following formula:

$$\mathrm{X}_i^{D1} = \mathrm{X}_i^{Tm} - \mathrm{X}_i^{Gm}. \tag{11}$$

Considering the difference feature $\mathrm{X}_i^{D1}$, its feature distribution is sparse feature units and does not have an overall feature distribution. Therefore, we use the fundamental spatial features $\mathrm{X}_i^{Tm}$ and $\mathrm{X}_i^{Gm}$ as the basis for feature distribution and enhance the $\mathrm{X}_i^{D1}$ feature through adaptive learning to reinforce specific information. Based on this, we highlight key information that distinguishes them by setting different functions that control information processing and filtering spatial features. The formula is expressed as follows:

$$\begin{cases} \mathrm{f}\left(x\right) = \dfrac{1}{1+e^{-x}} \\ \mathrm{g}\left(x\right) = \dfrac{1}{1+e^{-\tanh(x)}} \end{cases} \tag{12}$$

$$\mathrm{W}_{unique} = \text{Max}\Big(\text{Concat}\big(\mathrm{f}\left(\mathrm{X}_i^{D1}\right),\ \mathrm{g}\left(\mathrm{X}_i^{Tm}\right),\ \mathrm{g}\left(\mathrm{X}_i^{Gm}\right)\big)\Big) \tag{13}$$

$$\mathrm{X}_i^{U} = \mathrm{W}_{unique} \odot \left(\mathrm{X}_i^{T} + \mathrm{X}_i^{G}\right) \tag{14}$$

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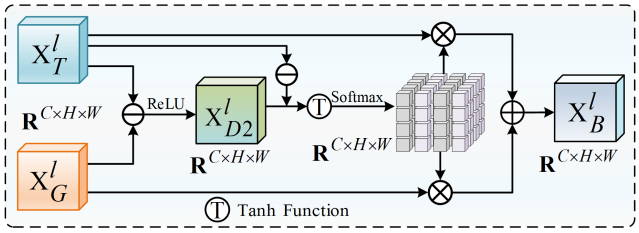



Fig. 4. Proposed differential basic features module. Basic information is derived through differential computation, and attention weights are generated via tanh nonlinear transformation followed by Softmax normalization. These attention weights are subsequently used to conduct a weighted fusion of the dual-branch feature maps, producing the differential base feature $X_i^B$.

where the functional distributions of $f(x)$ and $g(x)$ are utilized to amplify the proportion of differential feature weights by narrowing the range of fundamental spatial feature weights. Specifically, $f(x)$ maps differential responses to weights in $(0, 1)$, whereas $g(x)$ maps the original-branch responses to a narrower range of approximately $(0.269, 0.731)$, preventing either branch from dominating the spatial weighting. $W_{unique}$ enhances the unique features $X_i^U \in \mathbb{R}^{C_i\times H_i\times W_i}$ by selecting the maximum response values spatially through $\text{Max}(\cdot)$ and applying them to the dual-branch features. In this operation, the upper bound of attention weights for basic spatial features is approximately 0.73. For differential spatial features, a larger difference leads to a weight value closer to 1. This effectively enhances the response intensity of detailed features.

Through these operations, we obtained unique features with enhanced detailed information. These features utilize a differential method to decouple the differential responses of detailed characteristics, highlighting unique information by setting different functions, thereby improving the learning of discriminative features. Simultaneously, the integration of three different feature maps through maximum value selection preserves the contextual relevance and overall feature distribution.

*3) Differential Basic Features:* Given that background noise exhibits a branch-dependent characteristic distribution, its manifestation and impact vary across different branches. To further mitigate the interference from background noise, we implement the following operations. The module structure is shown in Fig. 4.

First, we utilize the differing response levels within the dual-branch feature maps to select features with low responses from both the object and the background, thereby constructing a low-response feature map. The procedure is as follows:

$$X_i^{D2} = X_i^T - \text{ReLU}\left(X_i^T - X_i^G\right) \tag{15}$$

where the $\text{ReLU}(\cdot)$ activation function is $\max(x, 0)$, which filters out positive values. $X_i^{D2}$ is then obtained by subtraction. However, considering issues such as extreme values and distribution imbalance in the feature maps, the following processing is continued:

$$X_i^B = \text{SoftMax}\left(\tanh(X_i^{D2})\right) \odot \left(X_i^T + X_i^G\right) \tag{16}$$

where the *tanh* $\in [-1, 1]$ activation function is used to constrain the range of eigenvalues, while the softmax function dynamically allocates the importance of basic feature regions across the global spatial scope. This operation primarily weakens background interference. It does so by scaling F1 and recalibrating the response weights of object features. As a result, the feature disparity between the background and the foreground is significantly amplified.

We leverage the discrepant background noise to differentially enhance the basic feature information, focusing exclusively on object features that elicit consistent responses across both branches during feature extraction. This operation is designed to reduce noise interference from the background on the object while strengthening the correlations among local features.

*4) Feature Fusion:* We obtain unique features and basic features through the differential method, and also use the residual method to obtain the original dual-branch features. We use feature addition for the fusion operation, which is expressed as follows:

$$X_i^S = X_i^U + X_i^B + \left(X_i^T + X_i^G\right) \tag{17}$$

where the residual connection is incorporated to mitigate object information loss during differential operations and ensure the integrity of the global context of the object.

In general, the DDSF strategy employs differential decoupling to extract both salient unique features and basic features. This not only enhances the learning of discriminative object characteristics but also suppresses background noise interference. As a result, object detection performance is remarkably improved in remote sensing images with complex backgrounds.

### C. Multiscale Dynamic Spatial Context Modeling

Due to the clustered arrangement and susceptibility to occlusion of small objects in complex remote sensing scenarios, missed and false detections often occur, leading to poor detection performance. Based on this, we propose the MDSCM module, whose structure is illustrated in Fig. 5.

Mamba [25] uses selective scanning to build long-range dependencies. This allows the model to be aware of the global context and to capture subtle feature differences between objects effectively. As a result, it helps mitigate the challenges of detecting small objects. However, existing vision Mamba structures mainly focus on general global spatial modeling through scan-order design, which remains insufficient for dense and occluded small objects with multiscale variations and irregular layouts. Considering the input consists of multiscale feature maps and aiming further to enhance the handling of dense and occluded small objects, the architecture of the original Mamba module has been redesigned in the MDSCM module.

We substitute the original convolutional operations with MSDeformAttn to allow more flexible focus on key regions across different scales. By integrating four-directional cross-selective scanning with a gated aggregation mechanism, dynamic spatial context modeling of multiscale features is achieved. Furthermore, by removing the original multiplicative branch and introducing an additive MSDeformAttn branch, the synergy between different components is enhanced. This

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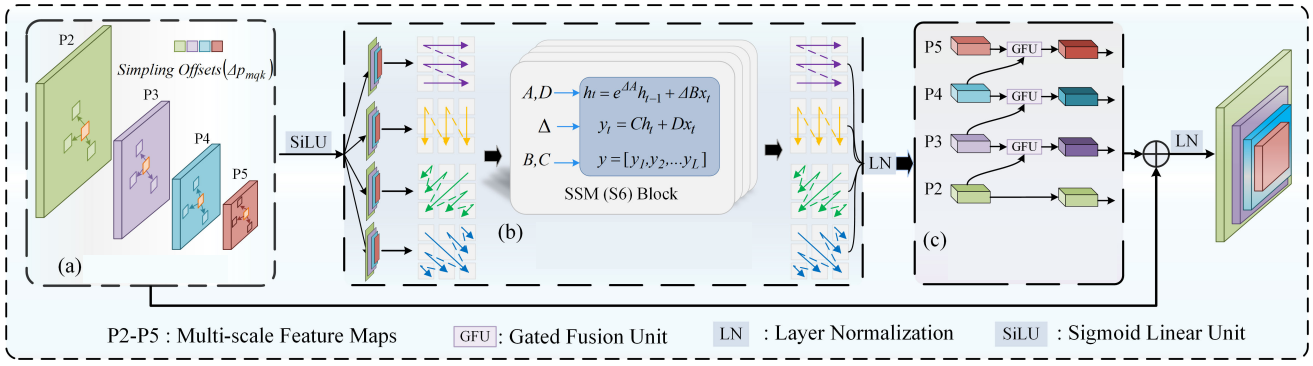



Fig. 5. Proposed MDSCM module. First, it utilizes (a) MSDeformAttn to extract local key information from multiscale features and dynamically samples multiscale interactions. Subsequently, dynamic parameter adjustment and four-directional global spatial modeling are performed in (b) four-directional cross-selective scan, followed by multiscale contextual interactions through (c) gated aggregation stream. Finally, the local and global information are further enhanced and fused for output.

allows the module to attend more accurately to small objects and their essential details within complex scenes. The key elements of the design are summarized below.

*1) Multiscale Deformable Attention:* To efficiently address multiscale features and small objects in remote sensing images, we employ MSDeformAttn for local feature extraction, which is formulated as follows:

$$\text{MSDeformAttn}(\mathcal{X}) = \sum_{m=1}^{M} W_m \left[ \sum_{l=1}^{L} \sum_{k=1}^{K} A_{mlqk} \cdot W'_m X_l \left( \hat{p}_q + \Delta p_{mlqk} \right) \right] \tag{18}$$

where $\mathcal{X} = \{X_l\}_{l=1}^{L}$ denotes the input multiscale feature maps. $\Delta p_{mlqk}$ and $A_{mlqk}$ represent the offset and attention weight of the $k$th sampling point in the $m$th attention head of the $l$th feature map, respectively. $\hat{p}_q$ is the normalized coordinate of the reference point for the $l$th feature map. $W_m \in \mathbb{R}^{d \times d}$ and $W'_m \in \mathbb{R}^{d \times d}$ are learnable weight matrices.

Through this operation, the fixed geometric structure of convolution is transformed into a dynamically adjustable receptive field attention mechanism, thereby enhancing the model's adaptability to irregular objects. Simultaneously, the parallel design of multiscale attention heads captures contextual information from different hierarchical levels, improving sensitivity in small object detection.

*2) Four-Directional Cross Selective Scan:* Unlike conventional vision Mamba scanning strategies that mainly emphasize global sequence modeling, we employ a four-directional cross-selective scan to better adapt the SSM block to visual positional information and spatial structures. By modeling multiscale feature maps from complementary directions, this design preserves 2-D spatial layouts and captures oblique or complex structures in remote sensing images, thereby improving the discrimination of dense and occluded small objects. The detailed architecture is shown in Fig. 5(b), followed by the operations described below:

Compared with unidirectional or bidirectional scanning along a single traversal axis, the four-directional scan captures complementary horizontal and vertical dependencies in both forward and reverse orders, thereby reducing directional bias in dense and occluded scenes

$$x_{tls} = \mathcal{S}_s(X_l), \quad l, s \in \{1, 2, 3, 4\} \tag{19}$$

$$Y_l = \sum_{s=1}^{4} \mathcal{U}_s\left(SSM(X_{tls})\right) \tag{20}$$

where $s \in \{1, 2, 3, 4\}$ denotes left-to-right, right-to-left, top-to-bottom, and bottom-to-top traversal, respectively. $\mathcal{S}_s(\cdot)$ deterministically rearranges $X_l$ into a channel-preserving 1-D sequence $X_{tls}$ of length $H_l W_l$. The $SSM(\cdot)$ (i.e., S6 block) performs selective state space modeling on each directional sequence $X_{tls}$. $\mathcal{U}_s(\cdot)$ is its inverse and restores each sequence element to its original 2-D coordinate.

This operation establishes nonlocal contextual correlations, enabling effective distinction between adjacent objects and completion of missing object parts. Leveraging the dynamic parameter adaptation mechanism of SSM, it suppresses redundant information while focusing on critical features. Furthermore, to enhance the detection of dense or occluded small objects in complex scenes, the multidirectional scanning strategy integrates contextual information from multiple perspectives. It effectively leverages the complementary relationships among them.

*3) Gated Aggregation Stream:* To further integrate the multiscale feature interactions of the four-directional cross SS2D, we designed a gated aggregation stream to fuse shallow detail features with deep semantic features, thereby enhancing multiscale perception. This process employs a bottom–up operation for multiscale fusion, as illustrated in the structure of Fig. 5(c). The core interactive unit, the designed gated fusion unit (GFU), performs aggregation operations for feature selection and dynamic fusion of multiscale features. Its detailed structure is shown in Fig. 6. The computational operation of GFU is expressed as follows:

$$X'_{l+1} = \sigma\left(\text{Mean}\left(\text{conv}_3(X_l)\right)\right) \odot X_{l+1} \tag{21}$$

$$X^G_{l+1} = \text{conv}_1\left(\text{Concat}\left(X'_{l+1}, X_{l+1}\right)\right). \tag{22}$$

In Fig. 6, $X_l$ and $X_{l+1}$ denote the low- and high-level features, respectively. Here, $\text{conv}_3$ denotes the $3 \times 3$ convolution that downsamples $X_l$ to the spatial resolution of $X_{l+1}$. Mean$(\cdot)$

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TABLE I
NUMBER OF OBJECTS PER CATEGORY IN NWPU VHR-10, DIOR, AND VEDAI

| Index | NWPU VHR-10 | | DIOR | | | | VEDAI | |
|---|---|---|---|---|---|---|---|---|
| | Category | Number | Category | Number | Category | Number | Category | Number |
| 1 | airplane | 757 | airplane | 1387 | ground-track-field | 2312 | boat | 170 |
| 2 | baseball-diamond | 390 | airport | 1310 | harbor | 1474 | camping-car | 390 |
| 3 | basketball-court | 159 | baseball-field | 2440 | overpass | 2019 | car | 1340 |
| 4 | bridge | 124 | basketball-court | 1369 | ship | 2702 | pickup | 950 |
| 5 | ground-track-field | 163 | bridge | 2176 | stadium | 1470 | tractor | 190 |
| 6 | harbor | 224 | chimney | 854 | storage-tank | 1614 | truck | 300 |
| 7 | ship | 302 | dam | 986 | tennis-court | 2582 | vans | 100 |
| 8 | storage-tank | 655 | expressway-service-area | 1125 | train-station | 994 | others | 200 |
| 9 | tennis-court | 524 | expressway-toll-station | 1182 | vehicle | 6420 | – | – |
| 10 | vehicle | 477 | golf course | 946 | wind-mill | 1616 | – | – |

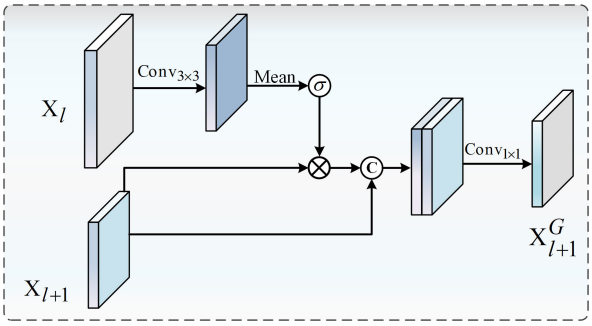



Fig. 6. Flowchart of the proposed GFU. Low-level features regulate high-level features via a gating mechanism, enabling cross-level interaction and thereby granting context-aware representations to deep features.

applies channel-wise averaging to form a single-channel spatial descriptor. The sigmoid function $\sigma$ generates a gating weight map, which is multiplied by $X_{l+1}$ to produce $X'_{l+1}$, allowing shallow details to guide high-level semantics. At the same time, high-level features focus on key regions, achieving detail-guided semantic enhancement. Finally, channel concatenation and compression correspond to Concat$(\cdot)$ and $\text{conv}_1$ in (22), respectively, and adaptively fuse complementary low-level details and high-level semantics.

The gated aggregation stream employs a bottom–up multiscale feature interaction mechanism. This mechanism synergizes with the multiscale feature fusion strategy of the MSDeformAttn module. Their features are then summed together. This process achieves efficient aggregation of cross-scale contextual features.

In summary, the MDSCM module is designed to improve detection performance on dense and occluded small objects substantially. The module explicitly models the structural priors and semantic relationships of small targets. It achieves this by dynamically focusing on multiscale local features, modeling multidirectional global spatial structures, and leveraging a gated aggregation stream for contextual interactions.

## IV. EXPERIMENTS

In this section, we first provide a brief introduction to the datasets used in the experiments. Subsequently, we present the specific experimental parameter configurations, accompanied by a comparative analysis of the utilized parameters. Finally, a series of ablation studies verifies the feasibility and effectiveness of the proposed modules, accompanied by thorough comparisons against other algorithms.

### A. Dataset Description

This article utilizes three datasets to evaluate the proposed method: NWPU VHR-10 [51], VEDAI [52], and DIOR [53].

*1) NWPU VHR-10:* NWPU VHR-10 is a ten-class geospatial object detection dataset released by NPU in 2014. The dataset consists of 800 remote sensing images in total—650 positive samples containing at least one object and 150 negative samples with no objects. It includes 715 color images from Google Earth with spatial resolutions ranging from 0.5 to 2 m, along with 85 pan-sharpened color infrared images from Vaihingen at a resolution of 0.08 m. Table I lists the class distribution, storage tank, and vehicles as small objects. Storage tanks typically exhibit dense arrangements, while vehicles appear in complex environments, enabling effective validation of our module.

*2) DIOR:* DIOR is a large-scale remote sensing object detection dataset containing 23 463 images and 190 288 object instances across 20 classes, with class-specific object counts detailed in Table I. The images, sourced from Google Earth, have a size of 800 $\times$ 800 pixels and spatial resolutions varying between 0.5 and 30 m. It exhibits significant variations in imaging conditions, climatic zones, seasonal changes, interclass similarity, and intraclass diversity.

*3) VEDAI:* VEDAI is a vehicle detection dataset comprising aerial image crops from the Utah AGRC dataset. It consists of two versions: 1024 $\times$ 1024 and 512 $\times$ 512 pixel images with spatial resolutions of 12.5 cm VEDAI_1024 and 25 cm VEDAI_512, respectively. Table I presents class categories and distributions, featuring extremely small vehicles and challenging factors such as multiorientation, specular reflection/occlusion, and illumination/shadow variations for robustness evaluation. The experiment adopts VEDAI_512 images, with the "plane" class excluded due to insufficient samples, resulting in an eight-class detection task.

### B. Experimental Setting

The baseline of our model uses Swin-tiny as the backbone. It incorporates the "look forward twice" and "mixed selection" components into the deformable-DETR [15] architecture, while utilizing pretrained weights on the ImageNet-1K dataset. The experiments are implemented in PyTorch on NVIDIA

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TABLE II
HYPERPARAMETER SETTINGS

| Hyperparameter | Value |
|---|---|
| Learning rate | 0.0001 |
| Weight decay | 0.0001 |
| Clip max norm | 0.1 |
| Encoder layer | 6 |
| Decoder layer | 6 |
| Number of query | 300 |
| Batch size | 2 |
| Dropout | 0.0 |
| Epoch | 36 |
| lr drop | 30 |

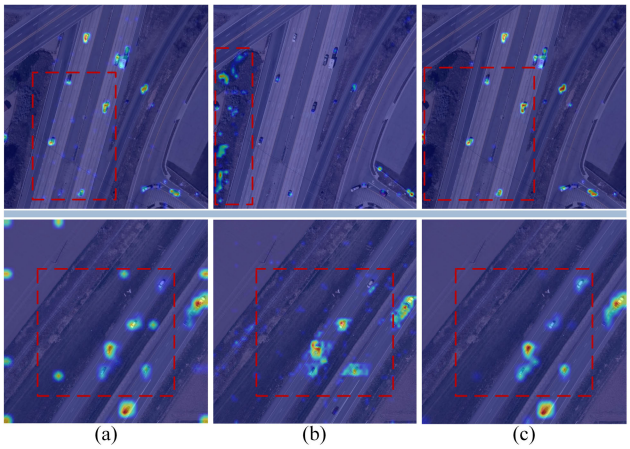



Fig. 7. Visualization of backbone feature heatmaps. The red boxes indicate responses to background noise. (a) Heatmaps produced by the SwinT branch. (b) Heatmaps produced by the GCNet branch. (c) Heatmaps after DDSF processing; compared with (a) and (b), DDSF suppresses background responses and enhances responses in object regions.

RTX 4090 GPUs with Ubuntu 22.04. We employ the AdamW optimizer with random horizontal flipping and cropping for data augmentation. Other training hyperparameters are set as shown in Table II. To ensure reliability, each experiment is conducted five times, and the results are averaged and reported. The loss function is configured following deformable-DETR: GIoU loss and $\ell_1$ loss are employed for bounding box regression, while focal loss is used for object classification, with respective weights of 2, 5, and 1.

### C. Ablation Study

In this section, we evaluate the performance of our proposed algorithm on the NWPU VHR-10, DIOR, and VEDAI datasets to demonstrate its effectiveness. The proposed innovations, including the DDSF strategy and MDSCM module, are integrated into the baseline network sequentially, and various AP metrics are used for evaluation ($AP_l$ is not evaluated on VEDAI due to the lack of large objects). To maintain a fair comparison, all experimental settings are kept identical except for the incorporation of the proposed method.

*1) Validation of DDSF Strategy:* The validity of our validation method depends on two main aspects: first, we visualize the heatmap of DDSF strategy features, as shown in Fig. 7. Second, we evaluate the experimental results based on multiple AP metrics, including:AP, $AP_{50}$, $AP_{75}$, $AP_s$, $AP_m$, and $AP_l$.

TABLE III
EXPERIMENTAL VERIFICATION OF THE PROPOSED MODULE ON NWPU VHR-10 (%)

| Baseline | DDSF | MDSCM | AP | $AP_{50}$ | $AP_{75}$ | $AP_s$ | $AP_m$ | $AP_l$ | #Params | GFLOPs |
|---|---|---|---|---|---|---|---|---|---|---|
| ✓ | × | × | 63.4 | 95.8 | 69.6 | 60.2 | 64.9 | 65.6 | 47.2M | 74.9 |
| ✓ | ✓ | × | 66.0 | 97.9 | 74.2 | 65.4 | **66.9** | 67.2 | 71.7M | 97.8 |
| ✓ | × | ✓ | 66.5 | 97.8 | 74.5 | 66.9 | 66.8 | 68.2 | 63.8M | 80.8 |
| ✓ | ✓ | ✓ | **67.0** | **98.5** | **76.5** | **67.5** | 66.0 | **69.1** | 88.3M | 103.7 |

TABLE IV
EXPERIMENTAL VERIFICATION OF THE PROPOSED MODULE ON DIOR (%)

| Baseline | DDSF | MDSCM | AP | $AP_{50}$ | $AP_{75}$ | $AP_s$ | $AP_m$ | $AP_l$ | #Params | GFLOPs |
|---|---|---|---|---|---|---|---|---|---|---|
| ✓ | × | × | 62.5 | 86.0 | 67.7 | 26.0 | 50.9 | 77.5 | 47.2M | 74.9 |
| ✓ | ✓ | × | 65.4 | 87.2 | 71.2 | 26.9 | 54.5 | 80.0 | 71.7M | 97.8 |
| ✓ | × | ✓ | 65.2 | 87.4 | 70.6 | 28.8 | 54.6 | 79.9 | 63.8M | 80.8 |
| ✓ | ✓ | ✓ | **66.7** | **88.3** | **72.9** | **30.7** | **58.5** | **81.5** | 88.3M | 103.7 |

TABLE V
EXPERIMENTAL VERIFICATION OF THE PROPOSED MODULE ON VEDAI (%)

| Baseline | DDSF | MDSCM | AP | $AP_{50}$ | $AP_{75}$ | $AP_s$ | $AP_m$ | #Params | GFLOPs |
|---|---|---|---|---|---|---|---|---|---|
| ✓ | × | × | 46.4 | 75.6 | 47.2 | 45.8 | 59.1 | 47.2M | 74.9 |
| ✓ | ✓ | × | 49.8 | 79.4 | 54.8 | 49.3 | 64.8 | 71.7M | 97.8 |
| ✓ | × | ✓ | 50.5 | 80.1 | 56.2 | 50.0 | 66.9 | 63.8M | 80.8 |
| ✓ | ✓ | ✓ | **51.4** | **81.8** | **58.3** | **51.7** | **67.5** | 88.3M | 103.7 |

TABLE VI
ABLATION STUDY ON THE EFFECTIVENESS OF THE DDSFS STRATEGY (%)

| Dataset | Fusion Strategy | AP | $AP_{50}$ | $AP_{75}$ | $AP_s$ | $AP_m$ | $AP_l$ |
|---|---|---|---|---|---|---|---|
| NWPU VHR-10 | Direct Add | 64.2 | 96.1 | 70.1 | 61.0 | 65.2 | 66.0 |
| | DDSF | **66.0** | **97.9** | **74.2** | **65.4** | **66.9** | **67.2** |
| DIOR | Direct Add | 63.2 | 86.4 | 68.9 | 26.3 | 52.1 | 79.3 |
| | DDSF | **65.4** | **87.2** | **71.2** | **26.9** | **54.5** | **80.0** |
| VEDAI | Direct Add | 47.1 | 78.0 | 49.8 | 45.8 | 60.8 | N/A |
| | DDSF | **49.8** | **79.4** | **54.8** | **49.3** | **64.8** | N/A |

Moreover, the number of stage blocks in the dual-branch backbone is (2, 2, 6, 2).

In Fig. 7, we can observe background interference and interclass interference in heatmaps (a) and (b) from the dual-branch backbone. However, in the heatmap (c), which is output by the DDSF strategy, the background noise is substantially weakened, and the high-response region exhibits clearer aggregated homogeneity. This proves the effectiveness of our DDSF strategy in the network.

From Tables III–V, we can also observe that when replacing the baseline backbone with a DDSF strategy, our network gets a significant improvement in AP metrics on all three datasets, especially on the VEDAI dataset where the AP improves by 3.4%, which indicates that the overall performance has been improved. The adaptability to complex environments has been enhanced.

To further demonstrate that the effectiveness of DDSF is not merely derived from dual-branch feature aggregation, we compare it with a naive feature addition strategy. As shown in Table VI, the direct summation of the SwinT and GCNet branch features is consistently inferior to DDSF across all

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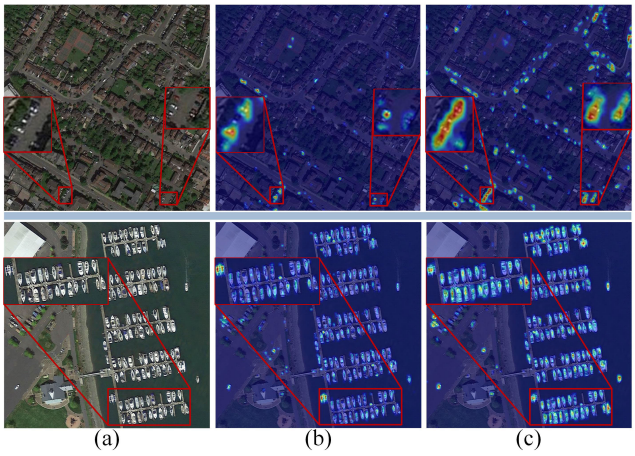



Fig. 8. Visualization of heatmaps of encoder layer feature maps. (a) Represents the original remote sensing image. (b) Represents a heatmap from the deformable-attention module encoder in the baseline network. (c) Represents heatmap from our MDSCM encoder, showing improved recognition of densely distributed small objects and precise localization.

three datasets. Compared to this simple fusion strategy, DDSF yields AP improvements of 1.8%, 2.2%, and 2.7% on NWPU VHR-10, DIOR, and VEDAI, respectively. This confirms that, instead of simply mixing features through direct addition, the proposed DDSF strategy effectively exploits global–local complementarity, enhances discriminative small-object features, and suppresses background interference.

*2) Validation of MDSCM Module:* In line with the DDSF validation strategy, we employ an MSDeformAttn module as the baseline. The effectiveness of this approach is validated using the attention heatmap visualized from the encoder.

In Fig. 8, we observe the first set of dense vehicle scenes (a). In these complex conditions, vehicles are difficult to distinguish due to occlusion or extremely small size. The heatmap (c) from our MDSCM encoder shows a significant improvement. This is evident when compared to the baseline's heatmap (b). It achieves much better detection performance for small vehicle objects. The second image contains a dense and occluded ship scene. Our result (c) can clearly distinguish and accurately locate every ship here. Meanwhile, the baseline's result (b) shows obvious thermal response fading and location shifts in the same area. This indicates that the baseline model has limited ability to capture intricate texture details. This comparison validates the superiority of the MDSCM module in object occlusion and dense small object detection tasks.

From Tables III–V, we observe a notable improvement in $AP_s$ across all three datasets compared to the baseline network: an increase of 6.7% on NWPU VHR-10, 2.8% on DIOR, and 4.2% on VEDAI.

Finally, to comprehensively evaluate the performance gain introduced by the proposed module, we integrated it into the baseline network. We then conducted comparative experiments, including detection visualizations and quantitative evaluations using AP metrics across three benchmark datasets. As shown in Figs. 9–11, the left figure represents the baseline network and the right figure is our proposed algorithmic network.

In the NWPU VHR-10 dataset (see Fig. 9), we can observe that the baseline network has significant limitations in detecting vehicles under occlusion conditions and is able to identify previously missed vehicles in our network. In the DIOR dataset (see Fig. 10), the baseline network, while effective in detecting large objects, performs poorly in detecting small objects such as oil storage tanks, boats, and small vehicles densely distributed in industrial areas. Compared to the missed detection of the baseline network, our network is able to detect densely distributed small objects successfully. In the VEDAI dataset (see Fig. 11), facing the challenges of high similarity between vehicle classes and complex environments, our network effectively suppresses false detections (e.g., vehicles in difficult scenarios such as residential areas) while maintaining high recall. Moreover, in Tables III– V, our method prioritizes AP metrics over the baseline network.

### D. Comparison With Other Methods

To further assess the advantages of the proposed method, we selected representative algorithms from recent years for comparative experiments, including SESANet [54], DINO [16], SED-DETR [55], Relation-DETR [56], Mamba-YOLO [57], AFSNet [58], SuperYOLO [59], PGCFNet [60], LSKNet [31], ESRTMDet [61], YOLOFusion [62], and FFCA-YOLO [63]. The experiments were conducted using the same parameter settings as listed in the model parameter table, and the comparative performance results are presented in Tables VII–IX.

On the NWPU VHR-10 dataset, we compare with the most advanced existing methods as shown in Table VII. In it, the mAPs of each category are included so that it is convenient to compare the detection effect of all categories.

SESANet uses the directional derivative model. Its adaptive derivative mask module suppresses large objects and focuses on small ones. The model also uses the sample importance assessment strategy, which solves the problem of traditional IoU loss ignoring small objects' prior information. As a result, it achieves higher AP for small objects like vehicles, airplanes, and storage tanks. However, this suppression of object interference may lose some object features, which can be observed in Table VII for basketball court and bridge, only 85.0% and 85.9%.

Relation-DETR improves object detection performance through several key innovations. It introduces an explicit position relation prior, employing an encoder to quantify box interactions and progressive attention to optimize cross-layer information. Additionally, it designs a contrast relation strategy that uses one-to-many matching to address prediction de-emphasis. However, in remote sensing images, it does not consider complex environments. It overlooks issues like small object occlusion and dense distribution. This leads to lower detection accuracy for ships and vehicles.

Mamba-YOLO uses a sequence modeling mechanism with linear complexity. It incorporates a multibranch structure, a gating mechanism, and depth-separable convolution. These components together improve its ability to model both local features and global dependencies for object detection. However, while increasing the sensing field, small objects become

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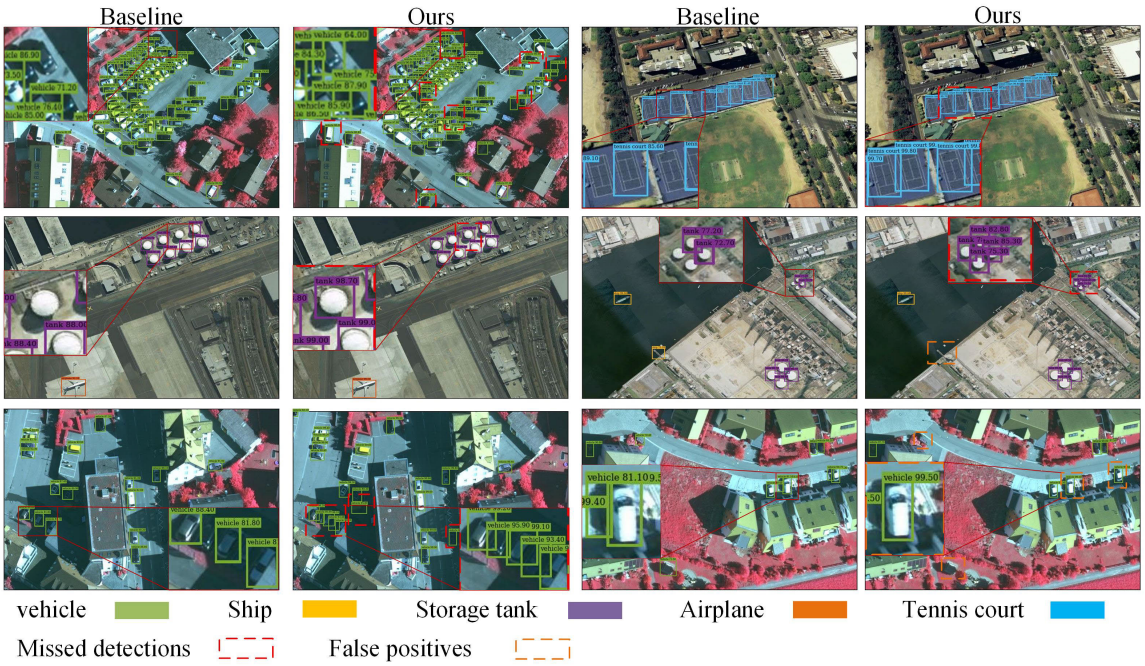



Fig. 9. Comparison of detection results on the NWPU VHR-10 dataset between the baseline network and the proposed D²M-DETR.

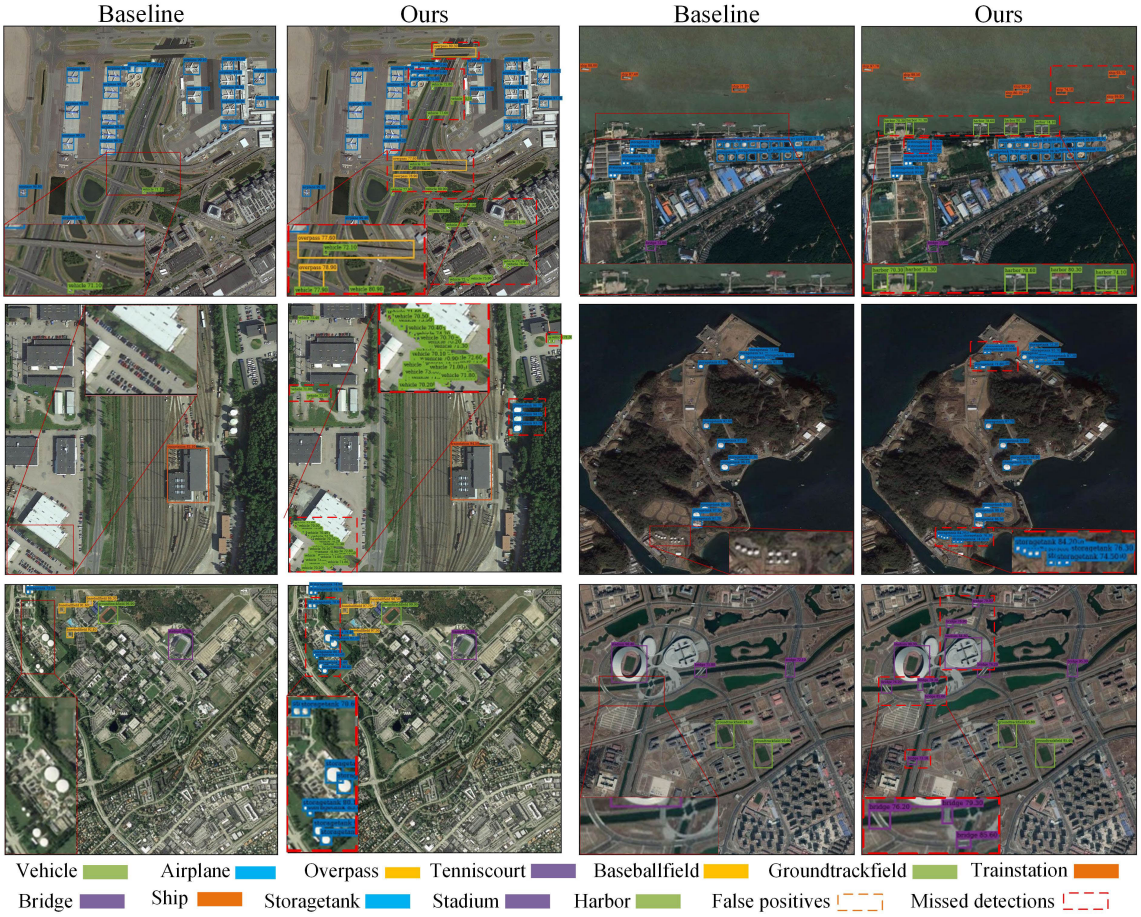



Fig. 10. Comparison of detection results on the DIOR dataset between the baseline network and the proposed D²M-DETR.

more vulnerable to complex environmental interference, leading to lower detection accuracy for categories such as vehicles.

AFSNet addresses the challenges of multiscale object detection and interference from complex backgrounds in remote

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Fig. 11. Comparison of detection results on the VEDAI dataset between the baseline network and the proposed D$^2$M-DETR.

TABLE VII

OBJECT DETECTION PERFORMANCE COMPARISON WITH OTHER METHODS ON NWPU VHR-10 (%)(AP AT IOU THRESHOLD 0.5)

| Methods | Airplane | Baseball diamond | Basketball court | Bridge | Ground track field | Harbor | Ship | Storage tank | Tennis court | Vehicle | mAP |
|---|---|---|---|---|---|---|---|---|---|---|---|
| SESANet [54] | **100.0** | 97.5 | 85.0 | 85.9 | 99.3 | 95.7 | 94.7 | 99.5 | 95.1 | 96.1 | 94.6 |
| AFSNet [58] | 98.4 | **100.0** | 99.8 | 98.2 | 98.3 | 97.5 | **97.0** | 99.9 | 93.6 | 94.3 | 97.7 |
| DINO [16] | 98.9 | 97.9 | 95.4 | 71.2 | 95.8 | 91.6 | 93.4 | 96.1 | 94.1 | 92.5 | 92.7 |
| SED-DETR [55] | 99.9 | 95.9 | 93.0 | 94.3 | 94.5 | 94.6 | 92.8 | 97.9 | 94.9 | 89.5 | 94.7 |
| Relation-DETR [56] | **100.0** | 96.9 | 91.0 | 93.2 | 97.0 | 99.1 | 91.9 | **100.0** | 95.2 | 92.8 | 95.7 |
| Mamba-YOLO [57] | 99.9 | 98.1 | 91.5 | 86.5 | 95.7 | 92.5 | 94.2 | 96.7 | 94.3 | 91.5 | 94.1 |
| PGCFNet [60] | 99.5 | 97.0 | 99.1 | 84.4 | 98.8 | 93.9 | 91.5 | 99.4 | 98.2 | 97.4 | 95.9 |
| Ours | 98.6 | 97.2 | **100.0** | **99.3** | **99.9** | **100.0** | 95.2 | 99.0 | **99.6** | **96.2** | **98.5** |

TABLE VIII

OBJECT DETECTION PERFORMANCE COMPARISON WITH OTHER METHODS ON DIOR (%)(AP AT IOU THRESHOLD 0.5)

| Methods | C1 | C2 | C3 | C4 | C5 | C6 | C7 | C8 | C9 | C10 | C11 | C12 | C13 | C14 | C15 | C16 | C17 | C18 | C19 | C20 | mAP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SESANet [54] | 93.5 | 78.2 | 91.9 | 82.2 | 46.8 | 89.3 | 61.4 | 67.2 | 67.5 | 73.3 | 76.7 | 55.8 | 62.0 | 78.8 | 84.8 | 82.4 | 89.8 | 54.4 | 76.7 | 79.1 | 74.6 |
| AFSNet [58] | 91.6 | 83.4 | 96.1 | 94.9 | 56.2 | 87.6 | 79.4 | 64.6 | 76.8 | 87.9 | 73.3 | 60.5 | 61.7 | 94.3 | 87.4 | 77.4 | 89.2 | 65.7 | 57.9 | 92.9 | 78.9 |
| DINO [16] | 87.7 | 90.8 | 85.6 | 91.1 | 57.6 | 84.8 | 79.3 | 93.8 | 81.1 | 86.2 | 86.2 | 64.2 | 69.5 | 78.9 | 80.0 | 73.7 | 91.8 | 70.1 | 61.0 | 91.5 | 80.2 |
| SED-DETR [55] | 85.0 | 92.5 | 86.3 | 91.0 | 55.5 | 84.6 | 77.2 | 92.8 | 84.0 | 86.5 | 86.0 | 67.1 | 67.8 | 79.8 | 80.5 | 74.5 | 92.9 | 71.7 | 59.0 | 90.4 | 80.3 |
| Relation-DETR [56] | 86.0 | 91.7 | 86.1 | 91.4 | 58.6 | 85.2 | 78.3 | 92.3 | 82.4 | 85.6 | 86.9 | 68.8 | 68.2 | 80.3 | 82.3 | 76.9 | 92.0 | 68.0 | 63.8 | 90.8 | 80.8 |
| Mamba-YOLO [57] | 92.7 | 82.6 | 85.4 | 91.7 | 48.1 | 82.9 | 64.1 | 76.0 | 72.4 | 84.2 | 81.7 | 67.9 | 64.2 | 92.4 | 81.6 | 83.0 | 92.8 | 66.6 | 59.0 | 84.0 | 77.7 |
| PGCFNet [60] | 97.7 | 90.7 | 94.3 | 91.0 | 59.0 | 90.0 | 76.3 | 94.6 | 79.6 | 87.4 | 91.0 | **74.4** | 77.9 | **96.3** | 95.0 | **89.3** | **97.0** | 62.2 | 70.0 | 95.7 | 86.0 |
| Ours | **98.8** | **96.3** | **96.5** | **96.2** | **71.5** | **92.6** | **85.3** | **95.9** | **85.8** | **90.5** | **93.7** | 70.7 | **81.4** | 90.3 | **95.8** | 83.1 | 96.0 | **74.6** | **74.9** | **96.6** | **88.3** |

sensing imagery. It proposes a hierarchical feature optimization and category-aware separation strategy. This strategy suppresses interference from large objects and enhances detail modeling for small objects. As a result, it effectively reduces intercategory interference and the impact of background noise. On the NWPU VHR-10 dataset, our method achieves the highest overall mAP, while AFSNet obtains the best AP values for the baseball-diamond and ship categories.

On the DIOR dataset, we compare with the most advanced existing methods as shown in Table VIII. In this case, we have carried out a standardized renaming for ease of observation. The objects corresponding to C1-C20 are as follows: airplane,

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TABLE IX
OBJECT DETECTION PERFORMANCE COMPARISON WITH OTHER METHODS ON VEDAI (%)(AP AT IOU THRESHOLD 0.5)

| Methods | Car | Pickup | Camping car | Truck | Other | Tractor | Boat | Van | mAP |
|---|---|---|---|---|---|---|---|---|---|
| LSKNet [31] | 83.0 | 77.5 | **85.0** | 53.9 | 45.6 | 79.4 | 55.6 | 57.4 | 67.2 |
| YOLOFusion [62] | 91.1 | 82.3 | 75.1 | 78.3 | 33.3 | 81.2 | 71.8 | 62.2 | 71.9 |
| SuperYOLO [59] | 90.3 | 82.7 | 76.7 | 68.6 | 53.9 | 79.5 | 58.1 | 70.3 | 72.5 |
| ESRTMDet [61] | **93.1** | **87.6** | 72.4 | 67.6 | 63.4 | **86.1** | 54.2 | **86.4** | 76.4 |
| FFCA-YOLO [63] | 89.6 | 85.7 | 78.7 | 85.7 | 48.6 | 81.8 | 64.5 | 67.0 | 74.8 |
| Mamba-YOLO [57] | 90.7 | 85.2 | 82.0 | 83.0 | **68.3** | 76.3 | 61.5 | 58.7 | 75.7 |
| DINO [16] | 88.0 | 85.6 | 83.0 | 79.5 | 63.6 | 70.6 | 84.4 | 67.4 | 77.8 |
| Ours | 91.4 | 81.1 | 84.7 | **87.7** | **68.3** | 84.9 | **81.6** | 75.0 | **81.8** |

airport, baseball-field, basketball-court, bridge, chimney, dam, expressway-service-area, expressway-toll-station, golf-course, ground-track-field, harbor, overpass, ship, stadium, storage-tank, tennis-court, train-station, vehicle, and windmill.

AFSNet, while performing well on the NWPU VHR-10 dataset, does not show good generalization on the DIOR dataset. Mamba-YOLO enhances local and global dependencies, but performs mediocrely on the three remotely sensed datasets in complex environments.

DINO uses contrast denoising training, dynamic anchor frame generation, and a hybrid query strategy. These methods solve the DETR family's problems of slow convergence, poor query initialization, and insufficient multiscale feature fusion in object detection. Corresponding to DINO detection results in these three datasets, it shows better generalization. However, no targeted processing is done for complex environments and dense small objects in remote sensing images.

SED-DETR solves three problems: multiscale objects, complex background interference, and poor training convergence. It uses a multiscale expansion sampling and adaptive scale aggregation strategy. It dynamically adjusts the sensory field to capture features of objects at different scales. It also uses deformable offsets to improve the modeled object shape and enhance fine-grained feature extraction. However, lacking the enhancement of dense small object context modeling capability, it can be observed that vehicle (59.0%) and ship (79.8%) detection accuracy is relatively low on the DIOR dataset.

PGCFNet enhances feature differentiation by using multiscale feature extraction and a dynamic channel reweighting mechanism. It combines a priori information to suppress false objects. It further enhances the discrimination of densely aligned small objects from the background by leveraging contextual modeling and optimizing multiscale feature aggregation. This approach solves the background interference problem caused by weak texture features and dense object alignment. The highest AP values are achieved for ships, storage tanks, and harbors on dense small objects. However, there are still limitations in the balance between large object feature retention and interference suppression, and the detection performance of some large objects is relatively low.

On the VEDAI dataset, we compare it with the most advanced existing methods as shown in Table IX. In this case, the dataset contains seven object categories about vehicles, and one object category is boat, which can be prone to problems such as misdetection and omission.

LSKNet tackles large-scale variations and complex backgrounds in remote sensing images by expanding receptive fields and refining feature extraction in key regions. It designs a mechanism to adjust the convolution kernel size dynamically. Additionally, it combines spatial and channel attention strategies to enhance multiscale object features while suppressing irrelevant interference. However, LSKNet shows some limitations in interclass interference suppression in scenes with similar objects. Camping car has significant features and can achieve the highest AP, but the detection accuracy of car and pickup is low.

ESRTMDet improves small object detection accuracy by introducing a lightweight super-resolution module and a feature enhancement module. It optimizes the training of the super-resolution branch by incorporating sensitive region weights (RoI weights). It removes the super-resolution module during inference to preserve the original detection network structure. As a result, it achieves the highest AP values on car, tractor, and van objects. However, due to the failure to process the complex environment, the sensitivity to some objects is weak, and lower AP values appear in boat and truck.

A thorough comparison of the algorithms above demonstrates that our network attains the highest mAP across all three datasets. It also achieves better AP values for multiscale objects and dense small objects in complex contexts across different datasets. In particular, our method enhances the accuracy of small objects without compromising the performance on large objects, demonstrating its effectiveness and superiority.

## V. CONCLUSION

This article addresses the dual challenges faced by small targets in complex remote sensing scenes: feature vulnerability to background interference and dense, occluded distribution. To tackle these issues, we propose an end-to-end network named D²M-DETR, based on differential decoupling and multiscale dynamic modeling. The effectiveness of the proposed method is demonstrated through experiments on multiple remote sensing datasets. The DDSF strategy suppresses background noise interference on small targets by leveraging response differences. This approach extends the perspective of feature disentanglement to address the issue of feature ambiguity for small targets in complex backgrounds. The MDSCM module promotes the deep integration of multidirectional state space modeling and multiscale attention mechanisms in remote sensing small object detection. It offers a novel approach to

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modeling structural priors and capturing cross-scale semantic associations of small objects in densely occluded scenarios. However, compared to the baseline, the dual-branch backbone and multiscale spatial modeling substantially raise both the computational complexity and the number of parameters in the model. Future work will focus on lightweight design, exploring strategies such as knowledge distillation and pruning to balance accuracy and efficiency better.

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